SOURCE: MSC.33(63).pdf
Hygroscopicit.y;
Drying of air and other gases;
Dew point; and
Low temperature effects .
.5 Health hazards
.5.1
.5.2
. 5 .3
Toxicity
.5.1.1
.5.1.2
.5.1.3
5.1.4
Hodes by which lique fi ed gases and their vapours may be toxic;
Toxic properties of inhibitors and of pro ducts, of combustion of both materials of construction and of liquefied gases carried;
Acute and chronlc effe c ts of toxicity. systemic poisons and irritants;
Threshold limit value (TLV) .
Hazards of skin contact, inha lation and ingestion .
First aid and administering of antidotes .
. 6 Cargo contaiDment.
.6.1
.6.2
.6.3
.6.4
6.5
Principles of containment systems;
Rules;
Surveys;
Tank construction, materials, coatings, insulation; and
Compatibility .
.7 Pollution
.7.1
. 7.2
.7.3
. 7.4
Hazards to human life and to the marine environment;
Effect of specific gravity and solUbility;
Danger from vapour cloud drift; and
Jettisoning of cryogenic liquids .
.8 Cargo handling system
.8.1
.8.2
.8.3
.8.4
.8.5
.8.6
.8.7
.8.8

.8.9
.8.10
.8.11
.8.12
Description of main types of pumps and pumping arrangements
and vapour return systems, piping systems and va l ves;

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SOURCE: MSC.33(63).pdf
.8 Cargo handling system
.8.1
.8.2
.8.3
.8.4
.8.5
.8.6
.8.7
.8.8
.8.9
.8.10
.8.11
.8.12
Description of main types of pumps and pumping arrangements
and vapour return systems, piping systems and va l ves;
Ezplanation of pressure, vacuum, suction, flow, head;
Filters and strainers;
Ezpansion devices;
Flame screens;
Commonly used inert gases;
Storage, gene ration, distributi on systems;
Temperature and pressure monitoring syst ems;
Cargo vent sys tems;
Liquid re-cir culation and re-liquefaction systems;
Cargo gauging, instrumentation sys tem s and alarms ;
Gas detection and monitoring systems;
RESOLUTION MSC.33(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CONVENTION
ON STANDARDS OF TRAINING, CERTIFICATION AND
WATCHKEEPING FOR SEAFARERS, 1978 - 9 -
. 8.13 C02 monitoring systems;
.8.14 Cargo boil-off systems: and
.8.15 Auxiliary systems
. 9 Ship operating procedures
.9.1
g. 2
.9.3
.9.4
g. 5
.9.6
.9.7
.9.8
.9.9

.9.10
Loading and disCharging preparat i ons and procedur es:
Check lists;
Cargo condition maintenance on passage and in harb our ;
Segregation of cargoe s and procedur es for cargo transfer;

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SOURCE: MSC.33(63).pdf
g. 2
.9.3
.9.4
g. 5
.9.6
.9.7
.9.8
.9.9
.9.10
Loading and disCharging preparat i ons and procedur es:
Check lists;
Cargo condition maintenance on passage and in harb our ;
Segregation of cargoe s and procedur es for cargo transfer;
Changing carg oes . tank cleaning procedures;
Cargo sampling;
Ballasting and de-ballasting;
Warm up and cool down systems:
Warm up and gas-freeing procedures; and
Procedures for cool down of gaS-free system from ambient
temperature and safety pre ca utio ns i n vo lved .
. 1 0 Safety practices and equipment
.10.1 Function. calibrat i on and use of portabl e measuring
ins trwnent s;
. 10 .2 Fi r e-fightin g equipm ent and procedures;
.10.3 Breathing apparatus;
.10.4 Resuscitators;
.10. 5 Escape se t s;
. 10.6 Rescue equipment;
.10.7 Protective clothing and equi pmen t:
. 10.8 Entry into enclosed spac es;
.10.9 Precauti ons to be observed before and during repair and
maintenan c e of cargo and control systems;
.10.10 Supervision of personnel during potentially hazardou s
operations:
.10.11 Types and principle s o f ce rtified safe electr ical equipmen t ;
and
.10 . 12 Sources of igniti on .

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SOURCE: MSC.33(63).pdf
maintenan c e of cargo and control systems;
.10.10 Supervision of personnel during potentially hazardou s
operations:
.10.11 Types and principle s o f ce rtified safe electr ical equipmen t ;

and
.10 . 12 Sources of igniti on .
. 11 Eaerqency procedures
.11.1
. 11.2
.ll .3
.ll.4
.11.5
The importance of developing ships' emergency plan s;
Emergency shutdown of cargo operations;
Emergen cy cargo valve closing systems:
Action in the event of failure o f systems or services essential to cargo; and
Action following collision or stranding. spillage and envelopmen t of ship in toxi c or flammable v a pour .
12 General principles of cargo operations
.12.1
.12.2
.12.3
.12 . 4
.12. 5
Inerting cargo tank and void spaces;
Tank cool down, loading;